

Medvale Internal Medicine

Chapter 2H - Sleep-Disordered Breathing, Obesity Hypoventilation, and Chronic Respiratory Failure

How to use this chapter

This chapter is original Medvale content for the pulmonary module. Uploaded pulmonary references and Anki-style signals are used only as coverage checklists. The text emphasizes clinical reasoning, numeric thresholds, and concept relationships for future graph RAG extraction.

Why this matters clinically

Sleep-disordered breathing is not simply a sleep clinic problem. In internal medicine, obstructive sleep apnea, central sleep apnea, and obesity hypoventilation syndrome often appear as unexplained fatigue, morning headache, resistant hypertension, atrial fibrillation, pulmonary hypertension, right-sided heart failure, chronic respiratory acidosis, or postoperative respiratory decompensation. A patient may not volunteer snoring or witnessed apneas unless asked directly, and a patient with chronic hypercapnia may look deceptively stable until an infection, sedative, opioid, heart failure exacerbation, or excessive oxygen administration pushes compensation into acute respiratory failure.

The organizing question is whether the patient primarily has an upper-airway obstruction problem, a central respiratory drive problem, a respiratory mechanics and obesity problem, or chronic gas-exchange failure from lung, chest wall, or neuromuscular disease. CPAP splints open a collapsible upper airway. BiPAP or noninvasive ventilation augments ventilation and CO₂ clearance. Oxygen improves hypoxemia but does not correct hypoventilation, and in CO₂ retainers oxygen must be titrated thoughtfully.

Core illness scripts

Syndrome	Typical patient	Key mechanism	High-yield clues	Diagnostic anchor	Management logic
Obstructive sleep apnea (OSA)	Middle-aged or older adult, often with obesity, large neck circumference, craniofacial crowding, or resistant hypertension	Recurrent upper-airway collapse during sleep despite ongoing respiratory effort	Loud snoring, witnessed apneas, gasping, morning headache, daytime sleepiness, nocturia, hypertension, AF	AHI ≥ 5 with symptoms or AHI ≥ 15 without symptoms; severity: 5-14 mild, 15-29 moderate, ≥ 30 severe	Weight loss, avoid alcohol/sedatives, PAP therapy, oral appliance or surgery in selected patients
Central sleep apnea (CSA)	Heart failure, stroke, opioid use, high altitude, neurologic disease, or idiopathic cases	Reduced or unstable central respiratory drive; airflow and effort fall together	Witnessed pauses may occur without snoring; Cheyne-Stokes crescendo-decrescendo pattern in HF	Sleep study shows central events; evaluate cardiac and neurologic context	Treat underlying cause; optimize HF; adjust opioids; sleep-specialist directed PAP/oxygen/other therapy
Obesity hypoventilation syndrome (OHS)	BMI ≥ 30 , usually severe obesity, often coexisting OSA	Increased mechanical load plus impaired ventilatory drive causes chronic daytime CO ₂ retention	Daytime somnolence, morning headache, dyspnea, edema, hypoxemia, high serum bicarbonate	BMI ≥ 30 + sleep-disordered breathing + awake PaCO ₂ ≥ 45 mmHg after excluding other causes	PAP therapy, weight loss, oxygen only if hypoxemia persists despite ventilation support
Chronic respiratory failure	COPD, ILD, neuromuscular disease, kyphoscoliosis, OHS, severe bronchiectasis	Long-term inability to maintain oxygenation, ventilation, or both	Low PaO ₂ , high PaCO ₂ , high HCO ₃ if chronic; acute worsening causes pH drop	Type 1: PaO ₂ < 60 . Type 2: PaCO ₂ > 45 . Acute-on-chronic: pH < 7.35 with chronic markers	Treat cause; oxygen targets; NIV for ventilatory failure; intubation if airway protection or work of breathing fails

Pathophysiology: oxygenation versus ventilation

A recurring mistake is to treat every low oxygen saturation as an oxygen problem. Oxygenation refers to movement of oxygen into arterial blood; ventilation refers to removal of carbon dioxide. A patient can be hypoxemic because of V/Q mismatch, shunt, diffusion limitation, low inspired oxygen, or hypoventilation. A patient becomes hypercapnic when alveolar ventilation is insufficient for CO₂ production. Hypercapnia therefore points toward hypoventilation, increased dead space, severe obstructive mechanics, neuromuscular weakness, CNS depression, or advanced exhaustion.

The alveolar-arterial gradient helps separate pure hypoventilation from lung parenchymal or vascular disease. Hypoventilation

from opioids, neuromuscular weakness, or obesity hypoventilation can produce hypoxemia with a normal A-a gradient. Pneumonia, pulmonary edema, pulmonary embolism, ARDS, and advanced COPD usually increase the A-a gradient because oxygen transfer is impaired by V/Q mismatch, diffusion abnormality, or shunt. Chronic hypercapnia generates renal compensation: the kidney retains bicarbonate over days, so a chronically elevated HCO_3^- can be a clue to long-standing CO_2 retention.

Original Diagram: ABG pattern logic in chronic respiratory failure



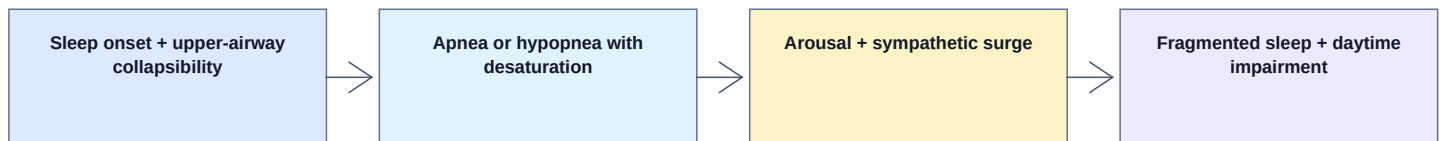
Normal A-a gradient with hypoxemia points toward hypoventilation or low inspired oxygen; elevated A-a gradient suggests V/Q mismatch, diffusion limitation, or shunt.

Obstructive sleep apnea

OSA is defined by repetitive collapse of the upper airway during sleep. The patient continues to make respiratory effort, but airflow falls or stops because the pharyngeal airway narrows. Each event can cause intermittent hypoxemia, hypercapnia, arousal, negative intrathoracic pressure swings, and sympathetic activation. Over months to years, these repeated events are associated with hypertension, atrial fibrillation, insulin resistance, pulmonary hypertension in susceptible patients, impaired concentration, motor vehicle crashes, and perioperative respiratory risk.

OSA should be suspected when a patient has loud snoring, witnessed apneas, choking or gasping at night, morning headaches, daytime sleepiness, resistant hypertension, unexplained polycythemia, atrial fibrillation, or obesity with a large neck circumference. Many patients underreport sleepiness because they normalize fatigue. A focused sleep history should ask about bed partner observations, nocturia, morning headaches, alcohol use before sleep, sedative use, opioid use, shift work, and occupational safety risk such as driving or operating equipment.

Original Diagram: The OSA cycle



Clinical outputs: loud snoring, witnessed apneas, morning headache, sleepiness, resistant hypertension, atrial fibrillation, pulmonary hypertension, and accident risk.

Diagnostic approach to OSA

The apnea-hypopnea index (AHI) counts apneas and hypopneas per hour of sleep. Common severity categories are mild OSA at 5-14 events/hour, moderate OSA at 15-29 events/hour, and severe OSA at ≥ 30 events/hour. A practical diagnostic threshold is $\text{AHI} \geq 5$ in a symptomatic patient or $\text{AHI} \geq 15$ even if symptoms are not prominent. Polysomnography is the classic test because it measures sleep stages, airflow, effort, oxygenation, and event type. Home sleep apnea testing can be appropriate for uncomplicated adults with a high pretest probability of moderate to severe OSA, but it is less suitable when central sleep apnea, hypoventilation, severe cardiopulmonary disease, neuromuscular weakness, chronic opioid use, or significant insomnia is suspected.

Questionnaires such as STOP-BANG are screening tools, not diagnostic tests. STOP-BANG assigns risk based on Snoring, Tiredness, Observed apnea, high blood Pressure, BMI, Age, Neck circumference, and Gender. A score of 3 or more increases suspicion, but a sleep test is needed to classify disease and guide therapy. Pulse oximetry alone is not enough because it cannot reliably distinguish obstructive events from central events or identify arousals without desaturation.

OSA data point	High-yield number or concept	Why it matters
AHI 5-14	Mild OSA	May still matter if sleepiness, hypertension, AF, or high occupational risk is present

AHI 15-29	Moderate OSA	Usually warrants active therapy when symptoms or cardiometabolic disease coexist
AHI ≥ 30	Severe OSA	Higher risk of hypoxemia burden, accidents, perioperative risk, and cardiopulmonary complications
STOP-BANG ≥ 3	Increased screening risk	Useful pretest tool; does not replace sleep testing
CPAP adherence benchmark	At least 4 hours/night on at least 70% of nights	Administrative and clinical benchmark; more nightly use usually gives more symptom benefit
Oxygen saturation nadir or time below 90%	Hypoxemia burden	Adds risk information beyond AHI, especially in COPD, OHS, and pulmonary hypertension

Management of OSA

Positive airway pressure is the main physiologic treatment for clinically important OSA because it splints the upper airway open during sleep. CPAP provides a constant pressure. Auto-titrating PAP adjusts pressure within a range. BiPAP provides separate inspiratory and expiratory pressures and is usually reserved for selected patients with pressure intolerance, hypoventilation, or coexisting ventilatory failure rather than routine uncomplicated OSA. The most common reasons for poor adherence are mask leak, nasal congestion, claustrophobia, dry mouth, pressure intolerance, and poor education at initiation. These barriers should be treated as fixable clinical problems, not as patient failure.

Weight loss improves OSA severity because less pharyngeal soft tissue and lower abdominal load reduce airway collapse and respiratory work. Alcohol and sedatives worsen OSA by reducing upper-airway muscle tone and arousal responses. Positional therapy may help patients whose events occur predominantly supine. Oral mandibular advancement devices can be useful for mild to moderate OSA or CPAP intolerance. Surgery is individualized; it is most effective when a correctable anatomic obstruction is identified, but it is not a universal substitute for PAP.

Central sleep apnea

Central sleep apnea differs from OSA because the problem is not physical obstruction of the upper airway. Instead, central respiratory drive becomes unstable or transiently absent. During a central apnea, both airflow and respiratory effort decrease. The classic internal medicine context is heart failure with Cheyne-Stokes breathing, where ventilation waxes and wanes in a crescendo-decrescendo pattern because circulation time, CO₂ feedback loops, and ventilatory control are unstable. Other causes include stroke, high altitude, chronic opioid therapy, renal failure, neurologic disease, and treatment-emergent central apnea after PAP therapy is started for OSA.

Management begins by treating the cause. In heart failure, optimization of guideline-directed heart failure therapy is central. In opioid-associated CSA, dose reduction or alternative pain strategies may be necessary. Supplemental oxygen or PAP may be used in selected phenotypes, but therapy should be individualized through sleep medicine because some modalities can be harmful in specific populations. A key example is adaptive servo-ventilation, which is generally avoided in patients with symptomatic heart failure with reduced ejection fraction and LVEF $\leq 45\%$ because of safety concerns identified in clinical trials.

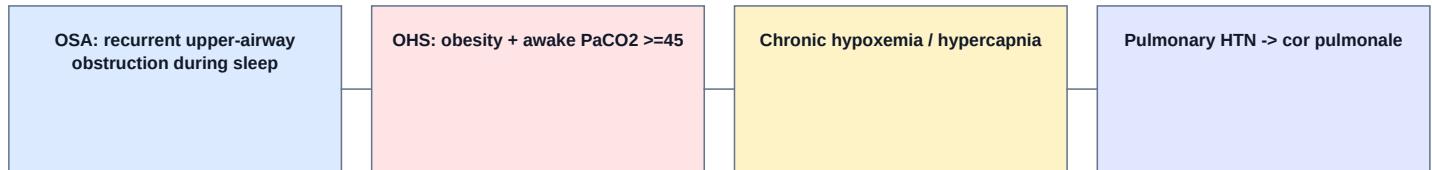
Feature	Obstructive sleep apnea	Central sleep apnea
Primary abnormality	Upper airway collapse	Loss or instability of central respiratory drive
Respiratory effort during event	Continues or increases against obstruction	Absent or reduced during central events
Common clues	Snoring, choking, obesity, large neck, witnessed obstruction	Heart failure, stroke, opioid use, high altitude, Cheyne-Stokes pattern
Sleep study pattern	Obstructive apneas/hypopneas, often with snoring and effort	Central apneas with reduced effort; may show crescendo-decrescendo ventilation
Treatment anchor	PAP to stent airway plus weight and risk factor management	Treat underlying cause; sleep-specialist directed PAP/oxygen/other therapy

Obesity hypoventilation syndrome

Obesity hypoventilation syndrome is diagnosed when three elements coexist: BMI ≥ 30 kg/m², sleep-disordered breathing, and awake daytime hypercapnia with PaCO₂ ≥ 45 mmHg after excluding other causes of hypoventilation. OHS is not the same as uncomplicated OSA. Many patients with OHS also have severe OSA, but OHS means the patient remains hypercapnic while awake. This implies that respiratory mechanics, ventilatory drive, or both have failed enough to cause chronic alveolar hypoventilation outside sleep.

The clinical picture often includes severe obesity, hypersomnolence, morning headache, dyspnea, edema, loud snoring, hypoxemia, erythrocytosis, pulmonary hypertension, and cor pulmonale. Serum bicarbonate is often elevated because the kidneys compensate for chronic respiratory acidosis. A bicarbonate level below about 27 mEq/L makes OHS less likely in a low-to-moderate suspicion setting, whereas a value ≥ 27 mEq/L should prompt consideration of ABG or venous/arterial CO₂ evaluation in the right clinical context. The definitive physiologic clue is awake hypercapnia.

Original Diagram: Sleep-disordered breathing to right-heart stress



Not every patient moves through the full chain, but persistent nocturnal hypoxemia and daytime hypercapnia increase pulmonary vascular load.

OHS management is ventilation-centered

The treatment goal in OHS is not merely to raise SpO₂ overnight. The goal is to improve ventilation, lower PaCO₂, prevent acute-on-chronic hypercapnic respiratory failure, and reduce pulmonary vascular stress. PAP therapy is first-line. CPAP is commonly used first when severe OSA is the dominant sleep phenotype, because splinting the airway may correct nocturnal obstruction and improve daytime CO₂ over time. BiPAP or NIV is favored when hypoventilation persists, when OSA is not severe, when CPAP fails, or when the patient is hospitalized with acute-on-chronic ventilatory failure. Oxygen may be added if hypoxemia persists after ventilation is addressed, but oxygen alone can worsen CO₂ retention because it does not fix hypoventilation.

Sustained weight loss is the disease-modifying intervention. Substantial weight loss, often in the range of 25-30% of body weight, is associated with improvement or resolution of hypoventilation in many patients. Bariatric surgery may be considered when lifestyle and medical therapy are insufficient and surgical risk is acceptable. Because untreated OHS carries high risk of hospitalization and cardiopulmonary complications, discharge planning after an acute episode should not stop at oxygen; it should include PAP/NIV access, follow-up sleep evaluation, ABG or CO₂ monitoring when appropriate, and weight management.

OHS concept	Number/threshold	Clinical interpretation
Obesity criterion	BMI ≥ 30 kg/m ²	Required for OHS definition, but most symptomatic patients have more severe obesity
Awake hypercapnia	PaCO ₂ ≥ 45 mmHg	Required; separates OHS from uncomplicated OSA
Serum bicarbonate clue	Often ≥ 27 mEq/L	Suggests chronic CO ₂ retention; not diagnostic alone
PAP approach	CPAP if severe OSA dominates; NIV/BiPAP if hypoventilation persists or OSA is not dominant	Treatment is based on physiology, not just oxygen saturation
Weight loss target for resolution	Often 25-30% body weight	Disease-modifying goal, especially with severe obesity
Oxygen caution	Avoid oxygen-only treatment when hypercapnia is present	Oxygen can improve saturation while ventilation and CO ₂ worsen

Chronic respiratory failure

Respiratory failure means the respiratory system cannot maintain adequate gas exchange for the body's metabolic needs. Type 1 respiratory failure is primarily hypoxemic, usually defined by PaO₂ < 60 mmHg on room air or a clinically equivalent oxygenation failure. Type 2 respiratory failure is hypercapnic, usually defined by PaCO₂ > 45 mmHg. Chronic respiratory failure develops when the abnormality persists long enough for physiologic adaptation, especially renal bicarbonate retention in chronic

hypercapnia. Acute-on-chronic respiratory failure occurs when a chronically compensated patient deteriorates and becomes acidemic, confused, severely dyspneic, or unable to protect the airway.

The most common chronic internal medicine settings are COPD, OHS, neuromuscular weakness, chest wall restriction, advanced ILD, bronchiectasis, and severe sleep-disordered breathing. COPD and OHS tend to produce hypercapnic failure when ventilatory mechanics or drive are insufficient. ILD more often produces hypoxemic failure because diffusion limitation and V/Q mismatch dominate, but late disease can also cause ventilatory failure through exhaustion or pulmonary hypertension. Neuromuscular disease causes ventilatory failure because respiratory muscles cannot generate adequate tidal volume; the lungs may be structurally normal.

Pattern	ABG profile	A-a gradient	Common causes	First management question
Pure hypoventilation	High PaCO ₂ , low PaO ₂ , pH depends on acute vs chronic state	Normal or near normal	Opioids, OHS, neuromuscular weakness, CNS depression	Does the patient need ventilation support rather than oxygen alone?
V/Q mismatch	Low PaO ₂ , PaCO ₂ low/normal early and high if severe	Elevated	COPD exacerbation, pneumonia, PE, pulmonary edema	Can oxygen fix it, and what is the underlying cause?
Shunt	Low PaO ₂ with poor response to oxygen	Elevated	ARDS, alveolar collapse, severe pneumonia, intracardiac shunt	Does the patient need PEEP, recruitment, or source control?
Diffusion limitation	Exertional hypoxemia, reduced DLCO	Elevated especially with exercise	ILD, emphysema, pulmonary vascular disease	Is this progressive parenchymal or vascular disease?
Chronic hypercapnia	High PaCO ₂ with high HCO ₃	Variable	COPD, OHS, neuromuscular/chest wall disease	Is compensation stable, or is pH falling?

Oxygen therapy: helpful, but not a ventilator

Oxygen therapy treats hypoxemia, not CO₂ retention. In chronic CO₂ retainers, especially COPD and OHS, oxygen should be titrated to a target that relieves dangerous hypoxemia while minimizing worsening hypercapnia. A common inpatient target is SpO₂ 88-92% for patients at risk of hypercapnic respiratory failure. The mechanism of oxygen-induced hypercapnia is multifactorial: worsened V/Q matching from loss of hypoxic pulmonary vasoconstriction, the Haldane effect reducing hemoglobin CO₂ carriage, and in some patients reduced ventilatory drive. The practical response is not to withhold oxygen from a hypoxemic patient; it is to titrate oxygen and check gas exchange when mental status, work of breathing, or CO₂ retention is concerning.

Long-term oxygen therapy improves survival in selected COPD patients with severe chronic resting hypoxemia. Classic criteria include PaO₂ ≤55 mmHg or SaO₂ ≤88% at rest, or PaO₂ 56-59 mmHg with evidence such as pulmonary hypertension, cor pulmonale, or erythrocytosis. Oxygen should be prescribed based on stable measurements rather than transient readings during an acute illness whenever possible. In contrast, oxygen for isolated exertional or nocturnal desaturation has more individualized benefit and should be considered in clinical context.

Oxygen scenario	Useful number	Interpretation
Acute COPD/OHS at risk of hypercapnia	Target SpO ₂ 88-92%	Titrate oxygen; check ABG/VBG if somnolence, acidosis, or rising CO ₂ is suspected
Most acutely ill patients not at hypercapnia risk	Often target SpO ₂ 92-96% or per institutional protocol	Avoid unnecessary hyperoxia but treat hypoxemia
Severe chronic resting hypoxemia in COPD	PaO ₂ ≤55 or SaO ₂ ≤88%	Long-term oxygen improves survival in appropriate patients
Borderline hypoxemia with right-heart strain	PaO ₂ 56-59 plus cor pulmonale, pulmonary HTN, or polycythemia	May qualify for long-term oxygen because hypoxemia is causing systemic effects
High FiO ₂ on ventilator	Prolonged FiO ₂ >60% raises oxygen toxicity concern	Increase PEEP or treat shunt physiology rather than relying indefinitely on high FiO ₂

Noninvasive ventilation and escalation

Noninvasive ventilation is most useful when the patient needs ventilatory assistance but can protect the airway and cooperate

with a mask. In acute COPD exacerbation with respiratory acidosis, BiPAP decreases work of breathing, improves alveolar ventilation, lowers PaCO₂, and can reduce intubation risk. It is also important in OHS-related acute-on-chronic hypercapnic respiratory failure. NIV is less reliable when the main problem is severe de novo hypoxemic respiratory failure, inability to clear secretions, vomiting, facial trauma, shock, or altered mental status that prevents airway protection.

A patient failing NIV should be recognized early. Warning signs include worsening pH, rising PaCO₂, persistent tachypnea, increasing accessory muscle use, inability to tolerate the mask, copious secretions, hemodynamic instability, hypoxemia despite support, or declining mental status. The next step is not to wait for exhaustion; it is to prepare for endotracheal intubation and invasive mechanical ventilation. In neuromuscular disease, declining vital capacity or negative inspiratory force may precede dramatic oxygen desaturation, so ventilation metrics and bulbar function matter.

Support modality	Main effect	Best use	Major limitation
Nasal cannula/simple mask	Raises inspired oxygen	Mild hypoxemia with stable ventilation	Does not reduce work of breathing or remove CO ₂
High-flow nasal cannula	High flow, some PEEP effect, dead-space washout, reliable FiO ₂	Hypoxemic respiratory failure, some CHF/pneumonia contexts	Less ventilatory support than BiPAP for severe hypercapnia
CPAP	Continuous airway pressure, airway splinting, recruitment	OSA, cardiogenic pulmonary edema, some hypoxemic states	Does not provide pressure support ventilation
BiPAP/NIV	Inspiratory pressure support plus expiratory pressure	COPD/OHS hypercapnic exacerbations, selected neuromuscular/chest wall disease	Requires airway protection, mask tolerance, manageable secretions
Invasive ventilation	Full airway control and ventilatory support	NIV failure, shock, airway compromise, severe altered mental status	Ventilator-associated risks, sedation, barotrauma, delirium

Presentation patterns and diagnostic approach

A practical bedside evaluation starts by sorting the patient's dominant problem. Sleepiness, snoring, and resistant hypertension point toward OSA. Severe obesity with daytime hypercapnia points toward OHS. Heart failure with crescendo-decrescendo breathing suggests CSA. Chronic dyspnea with hypoxemia and high bicarbonate suggests chronic respiratory failure with compensation. Acute confusion, somnolence, asterixis, headache, and respiratory acidosis suggest CO₂ narcosis or acute-on-chronic ventilatory failure. The oxygen saturation number alone is never enough; it must be paired with work of breathing, mental status, pH, PaCO₂, bicarbonate, and the underlying disease.

Testing should match the question. For suspected OSA in an otherwise uncomplicated patient, home sleep apnea testing may be sufficient. For suspected hypoventilation, severe cardiopulmonary disease, neuromuscular weakness, chronic opioid use, or central sleep apnea, in-lab polysomnography with CO₂ monitoring may be needed. For suspected OHS, obtain serum bicarbonate as a screening clue and ABG when suspicion is moderate to high. For chronic respiratory failure, ABG or VBG trend, pulmonary function testing, chest imaging, echocardiography for pulmonary hypertension/right-heart strain, and sleep testing may all be relevant depending on the phenotype.

Clinical clue	Likely implication	Next diagnostic move
Morning headaches + daytime sleepiness + loud snoring	Nocturnal hypoventilation or OSA	Sleep study; evaluate oxygen nadir and event type
BMI >=30 + awake PaCO ₂ >=45	OHS until another cause is proven	Exclude COPD, neuromuscular, chest wall, hypothyroid, drug causes; arrange PAP/NIV pathway
High serum bicarbonate in obese sleepy patient	Possible chronic CO ₂ retention	ABG/VBG and sleep evaluation if clinical suspicion is significant
Hypoxemia + normal A-a gradient	Hypoventilation or low inspired oxygen	Look for opioids, CNS depression, neuromuscular weakness, OHS
Hypoxemia + high A-a gradient	V/Q mismatch, shunt, diffusion, PE, edema, pneumonia	CXR/CT/PE workup/echo depending on context
Hypercapnia + low pH	Acute ventilatory failure or acute-on-chronic decompensation	NIV if appropriate; intubate if airway/mental status/shock/NIV failure

Complications and associated conditions

OSA and OHS create cardiovascular stress through sympathetic surges, intermittent hypoxemia, negative intrathoracic pressure swings, oxidative stress, and pulmonary vascular load. Hypertension is common and can be resistant. Atrial fibrillation can recur if OSA is untreated. Pulmonary hypertension and cor pulmonale are more likely when nocturnal hypoxemia is prolonged, when OHS coexists, or when chronic lung disease is present. Secondary erythrocytosis may occur when hypoxemia is sustained enough to raise erythropoietin drive. Neurocognitive impairment, mood symptoms, motor vehicle crashes, and reduced work performance may be the presenting burden rather than dyspnea.

In the hospital, sleep-disordered breathing becomes dangerous when sedatives, opioids, anesthesia, supine positioning, fluid overload, pneumonia, or COPD exacerbation reduce ventilatory reserve. A patient with undiagnosed OHS may be labeled as simply obese and sleepy when the actual problem is chronic ventilatory failure. After extubation, these patients may need immediate PAP/NIV support rather than routine oxygen alone. Discharge without a ventilation plan invites readmission.

Differentials and mimics

Mimic	How it resembles sleep-disordered breathing or chronic respiratory failure	Distinguishing clues
Hypothyroidism	Fatigue, weight gain, hypoventilation, macroglossia, sleepiness	High TSH/low free T4; can worsen OSA/OHS physiology
Depression or sedating medications	Sleepiness and poor concentration	Medication timing, mood symptoms, normal sleep study or additive sedative burden
Narcolepsy	Daytime sleepiness	Cataplexy, sleep paralysis, hypnagogic hallucinations; sleep latency testing
Heart failure	Nocturnal dyspnea, CSA, fatigue	Volume overload, low EF or diastolic dysfunction, Cheyne-Stokes respiration
COPD alone	Hypercapnia and dyspnea	Obstructive spirometry; may overlap with OSA as overlap syndrome
Neuromuscular weakness	Hypoventilation with normal lung parenchyma	Weak cough, dysphagia, low vital capacity, low maximal inspiratory pressure
Pulmonary embolism or pneumonia	Acute dyspnea/hypoxemia	Elevated A-a gradient, imaging/lab evidence, acute pleuritic/infectious context

High-yield exam clues

1. An obese patient with daytime somnolence and morning headaches but normal daytime PaCO₂ most likely has uncomplicated OSA. An obese patient with daytime hypercapnia, hypoxemia, elevated bicarbonate, and respiratory acidosis has OHS or another chronic hypoventilation syndrome until proven otherwise.
2. Hypoxemia with a normal A-a gradient points toward hypoventilation. Hypoxemia with an elevated A-a gradient points toward V/Q mismatch, diffusion limitation, shunt, pulmonary vascular disease, or parenchymal lung disease.
3. In COPD or OHS patients at risk for CO₂ retention, oxygen is still given when needed, but it is titrated. A common target is SpO₂ 88-92%, and worsening mental status or acidosis should prompt blood gas assessment and ventilatory support consideration.
4. CPAP treats upper-airway collapse. BiPAP/NIV treats ventilation failure by providing inspiratory pressure support. Oxygen treats hypoxemia but does not remove CO₂.
5. Prior respiratory failure is a major warning sign in asthma, COPD, OHS, neuromuscular disease, and sleep-related breathing disorders because it means the patient has already crossed the threshold from symptoms into ventilatory collapse under stress.

Common pitfalls

Pitfall 1: Assuming normal daytime oxygen saturation excludes OSA. Many OSA patients have normal daytime SpO₂ and abnormal nocturnal physiology. Pitfall 2: Treating OHS with oxygen alone. This may make the monitor look better while PaCO₂

and acidosis worsen. Pitfall 3: Calling a patient nonadherent to CPAP without troubleshooting mask fit, leak, nasal symptoms, pressure intolerance, and education. Pitfall 4: Using home sleep apnea testing in a patient with suspected central apnea, hypoventilation, severe cardiopulmonary disease, or neuromuscular weakness, where in-lab study is often more informative. Pitfall 5: Waiting too long to intubate a patient failing NIV.

Practice questions

Question 1

A 52-year-old man with BMI 42 has loud snoring, witnessed apneas, morning headaches, and daytime sleepiness. ABG while awake shows pH 7.39, PaCO₂ 41, PaO₂ 82. What is the most likely diagnosis?

Answer: Obstructive sleep apnea. His symptoms strongly suggest OSA, and his normal awake PaCO₂ argues against obesity hypoventilation syndrome. A sleep study is the diagnostic next step.

Question 2

A 58-year-old woman with BMI 51 presents with somnolence and dyspnea. ABG shows pH 7.33, PaCO₂ 62, PaO₂ 58, HCO₃ 33. What syndrome should be suspected?

Answer: Obesity hypoventilation syndrome with acute-on-chronic hypercapnic respiratory failure. BMI ≥ 30 plus awake PaCO₂ ≥ 45 after excluding other causes defines OHS; elevated bicarbonate indicates chronic compensation, while low pH indicates acute decompensation.

Question 3

A patient with COPD exacerbation has SpO₂ 77% on room air. A trainee suggests avoiding oxygen because the patient may retain CO₂. What is the correct principle?

Answer: Give oxygen, but titrate it. In a COPD patient at risk for hypercapnia, a common target is SpO₂ 88-92%, with blood gas monitoring if mental status, acidosis, or rising CO₂ is a concern.

Question 4

A patient with heart failure has periodic crescendo-decrescendo breathing and a sleep study showing apneas with absent respiratory effort. What category is this?

Answer: Central sleep apnea, specifically a Cheyne-Stokes pattern in the appropriate heart failure context. Treatment starts with optimizing the underlying heart failure and sleep-specialist directed therapy.

Question 5

A patient with chronic hypercapnic COPD is admitted with acute respiratory acidosis. He is awake, cooperative, and protecting his airway. What respiratory support is most physiologically targeted?

Answer: Noninvasive ventilation such as BiPAP, because it provides inspiratory pressure support, lowers work of breathing, and improves CO₂ clearance. Intubation is needed if NIV fails or airway protection is compromised.

Graph-RAG extraction map

Node	Useful aliases	Key attributes	High-value edges
Obstructive sleep apnea	OSA; upper-airway collapse	AHI categories, snoring, witnessed apnea, CPAP, hypertension/AF links	causes intermittent hypoxemia; treated by CPAP; associated with resistant hypertension; overlaps with COPD/OHS
Central sleep apnea	CSA; Cheyne-Stokes breathing	Absent effort, HF/opioids/stroke/high altitude, central events	caused by heart failure; differentiated from OSA by respiratory effort; may be worsened by opioids
Obesity hypoventilation syndrome	OHS; Pickwickian physiology	BMI ≥ 30 , awake PaCO ₂ ≥ 45 , HCO ₃ often high, PAP/NIV, weight loss	causes chronic hypercapnia; overlaps with OSA; may cause pulmonary hypertension and cor pulmonale
Chronic respiratory failure	Type 1; type 2; acute-on-chronic	PaO ₂ < 60 , PaCO ₂ > 45 , pH/HCO ₃ patterns, oxygen/NIV/intubation	caused by COPD/OHS/ILD/neuromuscular disease; managed by oxygen or ventilation depending on physiology
Noninvasive ventilation	BiPAP; NIPPV; NIV	IPAP/EPAP concept, acute COPD/OHS, airway protection needed	treats hypercapnic respiratory failure; fails with shock/AMS/secretions; alternative to intubation in selected patients

Long-term oxygen therapy	LTOT; home oxygen	PaO ₂ ≤55 or SaO ₂ ≤88; PaO ₂ 56-59 with cor pulmonale/polycythemia	treats severe chronic hypoxemia; does not treat hypoventilation; may be combined with NIV
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Selected reference anchors used for clinical cross-checking

American Academy of Sleep Medicine clinical practice guidelines for diagnostic testing and positive airway pressure treatment of adult obstructive sleep apnea; American Thoracic Society clinical practice guideline on obesity hypoventilation syndrome; American Thoracic Society guideline on long-term noninvasive ventilation in chronic stable hypercapnic COPD; British Thoracic Society oxygen guidance and GOLD COPD materials for oxygen targets and chronic hypoxemia concepts; standard pulmonary/critical care references for respiratory failure physiology.

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